

Lauren Holian

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RESEARCH INTERESTS

Disease ecology; spatial and temporal community dynamics; host-parasite interactions; abundance-occupancy relationships; landscape change; zoonotic disease risk; conservation

EDUCATION

PhD in Biological Sciences, University of South Carolina Expected: Fall 2026

Advisor: Dr. Tad Dallas

Master of Science in Zoology, University of Florida 2019

Advisor: Dr. Jamie Gillooly

Bachelor of Science in Integrative Biology, University of Florida 2018

Honors Program and magna cum laude

Minor in Sustainability Studies

MANUSCRIPTS IN REVISION

- **Holian, L.A.**, Ten Caten, C., Dallas, T. Examining the biogeographic patterns of ground beetle community synchrony. In revision, *Journal of Animal Ecology*.

PEER-REVIEWED PUBLICATIONS

- **Holian, L.A.**, Ten Caten, C., Dallas, T. (2026). Spatial and temporal perspectives on occupancy and abundance. *Ecology & Evolution*.
- Dallas, T. A., **Holian, L. A.**, & Ten Caten, C. (2025). Landscape structure and species life history affect abundance-occupancy relationships. *Ecology*, 106(11), e70258.
- Ten Caten, C., **Holian, L. A.**, & Dallas, T. A. (2025). The influence of geographic ranges, climatic niches and temperature fluctuations on population variability. *Proceedings of the Royal Society B: Biological Sciences*, 292(2051).
- Dallas, T. A., **Holian, L. A.**, & Caten, C. T. (2024). Geographic and temporal distance–decay relationships across taxa. *Oikos*, e10269.
- Dallas, T. A., Caten, C. T., & **Holian, L. A.** (2023). Temporal variability of carabid beetles as a function of geography, environment, and species. *Theoretical Ecology*, 1-9.
- Ten Caten, C., **Holian, L. A.**, & Dallas, T. (2022). Effects of occupancy estimation on abundance–occupancy relationships. *Biology Letters*, 18(6), 20220137.
- Ten Caten, C., **Holian, L.A.**, Dallas, T. (2022). Weak but consistent abundance-occupancy relationships across taxa, space, and time. *Global Ecology and Biogeography*
- **Holian, L.A.**, Anderson, D.M, Gillooly, J.F. (2020). Host Cell Volume Explains Differences in the Size of DsDNA Viruses. *Virus Research*
- Powell, E. C., Cook, C., Coco, J., Brock, M., **Holian, L. A.**, & Taylor, L. A. (2019). Prey colour biases in jumping spiders (*Habronattus brunneus*) differ across populations. *Ethology*.
- Nakahara, S., Zacca, T., Huertas, B., Neild, A. F., Hall, J. P., Lamas, G., **Holian, L. A.**, Espeland, M., & Willmott, K. R. 2018. Revision of the ‘aegrota species group’ of *Caeruleptychia* Forster, 1964, with three new species. *Insect Systematics & Evolution*, 49(2), 130–182.

SCHOLARLY COMMENTARIES & PEDAGOGICAL PUBLICATIONS

- **Holian, L.A.**, Ten Caten, C., Dallas, T.A. (2023). Exploring species diversity across space and time with data from the National Ecological Observatory Network. Teaching Issues and Experiments in Ecology, Vol.19: Practice #10. https://tiee.esa.org/vol/v19/issues/data_sets/holian/abstract.html
- Richards, R.L. & **Holian, L.A.** (2022). Infectious disease: Dog diets may drive transmission cycles in human Guinea worm disease. *Current Biology* —Dispatch Article
- Dallas, T., **Holian, L.A.**, Foster, G. (2020). What determines parasite species richness across host species? *Journal of Animal Ecology* —In Focus Article

SELECTED RESEARCH EXPERIENCE

Dallas Lab, University of South Carolina

May 2021–Present

- Lead research on spatial and temporal patterns in abundance-occupancy relationships, community synchrony, and host-parasite interactions across large ecological datasets
- Collaborate on projects examining population variability, synchrony, and macroecological patterns across taxa, space, and time
- Developed and maintain *Tribolium castaneum* and *Daphnia dentifera* laboratory systems, including husbandry protocols, experimental design, and undergraduate mentorship

Climate & Population Variability Working Group, ESIIL

November 2025–Present

- Co-lead a project investigating the structure of synchrony networks across space and time
- Contribute to collaborative projects examining links among climate variability, population variability, and community synchrony

EcoHealthNet Research Exchange Program

May 2023–August 2023

Developed an ongoing collaboration examining how landscape composition and configuration affect zoonotic disease risk in Minas Gerais, Brazil

Gillooly Lab, University of Florida

August 2016–August 2019

- Led a manuscript evaluating virus-host size relationships for dsDNA viruses infecting the three domains of life using two measures of size (genome size and cell volume).
- Investigated viral decay rates across temperature gradients

Undergraduate Research, University of Florida

January 2014–August 2018

- Investigated sexual selection in jumping spiders (genus *Habronattus*)
- Produced morphological illustration of butterflies for species identification at the Florida Museum of Natural History

RESEARCH TRAINING AND FELLOWSHIPS

EEID Spatial epidemiological modeling workshop

May 2026–June 2026

Completed workshop on modeling spatially explicit epidemiological processes; simulated spatially explicit dengue outbreaks under various initial conditions

Fellow: NIH T35 Big Data Health Science Training Program

May 2024–August 2024

Completed training in epidemiology and AI/data-science methods; modeled dengue outbreak predictability in El Salvador using temperature data in collaboration with Melissa Nolan

Community fellow: Earth Science Information Partners

January 2023–December, 2024

Provided logistical and administrative support for the Open Science Cluster; engaged with the ESIP community; supported virtual and in-person annual meetings

LEAD AUTHOR ORAL PRESENTATIONS

Ecology & Evolution of Infectious Disease

Richness & compositional dissimilarity of helminth & viral communities across mammal hosts 2022

Ecological Society of America

Predicting helminth interactions in threatened mammalian hosts 2025

Spatial patterns of community synchrony 2024

Patterns of prevalence and intensity: How helminths are distributed across mammalian hosts 2022

Earth Science Information Partnership

Trade-offs & synergies in the planning of health landscapes 2024

International Association of Landscape Ecology Congress: Nairobi, Kenya

Trade-offs & synergies in the planning of health landscapes 2023

LEAD AUTHOR POSTER PRESENTATIONS

Ecology & Evolution of Infectious Disease

Known & predicted differences in parasite richness between mammalian threat classes 2026

Ecological Society of America

Interaction of spatial & temporal patterns in abundance-occupancy relationships 2021

MENTORSHIP EXPERIENCE

Dallas Lab, University of South Carolina

October 2022–Present

Recruited, trained, and supervised multiple undergraduate researchers working with *Tribolium castaneum* and *Daphnia dentifera*. Mentored students in project development, husbandry, data collection, internal grant proposals, and manuscript preparation; one project submitted for publication

Gillooly Lab, University of Florida

May 2019–August 2019

Recruited and assisted three undergraduate researchers on scientific literacy and data entry/analysis

FIELD/ANIMAL EXPERIENCE

Animal care (keeper), Zoo Miami

August 2019–August 2020

- Supported animal welfare and management through behavioral observation, recordkeeping, and husbandry in an AZA-accredited zoological setting
- Supported veterinary care, animal training, and conservation education for a diverse hoofstock collection, including black rhinoceros, greater one-horned rhinoceros, banteng, okapi, giraffe, antelope, equids, and pygmy hippopotamus

Native Trout Conservation, Yellowstone National Park

May–August 2018

Conducted backcountry stream surveys for native trout conservation, including electrofishing, eDNA collection, genetic sampling, invasive species removal, telemetry tracking, species identification, otolith removal, and data entry

Mammal trapping, PORTAL Ecological Project

October 2018 & January 2019

Assisted in rodent surveys; rodent trapping/handling, species ID, weighing, and sexing

Spider field sampling, University of Florida

January 2014–May 2016

Supported jumping spider (genus *Habronattus*) breeding, rearing, daily care, and field collection

TEACHING EXPERIENCE

Graduate Teaching Assistant 2018–2024

- Instructed undergraduate laboratory courses in ecology, biology, and interdisciplinary STEM topics
Led weekly lab sections for courses ranging from 1–3 credits and 16–48 students, for 4–6 hours per week
- Universities: University of South Carolina; Louisiana State University; University of Florida
- Courses: Ecology Lab; Biology Lab 1; Biology Lab 2; Interdisciplinary X-Lab 1 & 2

Curriculum development September 2022–August 2023

Co-developed an undergraduate lab that explores species diversity concepts through data manipulation and discussion; published in *Teaching Issues and Experiments in Ecology*

SCIENTIFIC AND CONSERVATIONS SERVICE

IUCN Commission for Ecosystem Management (Human Health) November 2023–Present

Contributed to the development of outreach materials and the development of the ecosystem index of human health services

Student liaison, Ecological Society of America; Disease Ecology section August 2025–Present

Elected to represent and convey the interests of graduate students to the section leadership

Peer review February 2024–Present

Served as a peer reviewer for *Ecology*, *Teaching Issues and Experiments In Ecology*, *Ecology Letters*

World Organization for Animal Health (WOAH) Report January 2023

Screened 2,000 abstracts for WOAH's literature review on wildlife diseases of conservation concern

CodeFest/MakerSpace November & December 2021

Supported students through a programming and hardware session while fostering creativity, teamwork, and introductions to data science

Invasive Cane Toad Collection, Zoo Miami August 2020

Assisted the annual nocturnal collection of invasive cane toads

Guest Speaker, Archbishop Coleman F. Carroll High School February 2020

Presented to 3 classes on biodiversity & conservation topics

Wetlands Club President, University of Florida August 2016–May 2018

Led student conservation service activities, including wetland restoration, SEEP wetland maintenance, community outreach, and BioBlitz participation

Professional membership: Ecological Society of America 2021–Present

HONORS & AWARDS

Elsie Taber Fellowship, University of South Carolina 2022, 2023

Graduate Travel Grant, University of South Carolina 2023

ESA Southeast Chapter Student Travel Award 2022

McDaniel Travel Award, Louisiana State University 2021

SKILLS

Languages: English and Spanish

Programming and reproducible research: R, Git/GitHub, reproducible workflows

Quantitative ecology: Experience with large datasets, spatial analyses, population/community modeling